Problem 1

1a

What is the first principal component w_1 (use the unbiased estimation of covariance)? Draw the first principal component direction w_1 on the plot, anchored at the origin.

$$X = \begin{bmatrix} 0 & -1 & -1 & -2 \\ 2 & 1 & 2 & 3 \end{bmatrix}$$

We calculate the mean

$$\mu = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

Then we center the data by subtracting the mean from each point in X

$$X_{centered} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & -1 & 0 & 1 \end{bmatrix}$$

The covariance is modeled by the equation $\frac{1}{n-1}XX^T$ where n is the number of data points.

$$cov = \frac{1}{3}\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \\ 0 & 0 \\ -1 & 1 \end{bmatrix}$$

$$cov = \begin{bmatrix} \frac{2}{3} & \frac{-1}{3} \\ \frac{-1}{3} & \frac{2}{3} \end{bmatrix}$$

To calculate the eigenvalues, we solve for λ in $|A - \lambda I| = 0$

$$\left| \begin{bmatrix} \frac{2}{3} - \lambda & \frac{-1}{3} \\ \frac{-1}{3} & \frac{2}{3} - \lambda \end{bmatrix} \right| = 0$$

$$\lambda^2 - \frac{4}{3}\lambda + \frac{4}{9} - \frac{1}{9} = 0$$

$$(\lambda - 1)(\lambda - \frac{1}{3}) = 0$$

$$\lambda_1 = 1$$

$$\lambda_2 = \frac{1}{3}$$

We then use λ_1 to calculate the corresponding eigenvector v_1

$$\begin{bmatrix} \frac{2}{3} - 1 & \frac{-1}{3} \\ \frac{-1}{3} & \frac{2}{3} - 1 \end{bmatrix} v_1 = 0$$

$$\begin{bmatrix} \frac{-1}{3} & \frac{-1}{3} \\ \frac{-1}{3} & \frac{-1}{3} \end{bmatrix} v_1 = 0$$

$$v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Finally, we normalize the eigenvector to get the first principle component.

$$w_1 = \begin{bmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

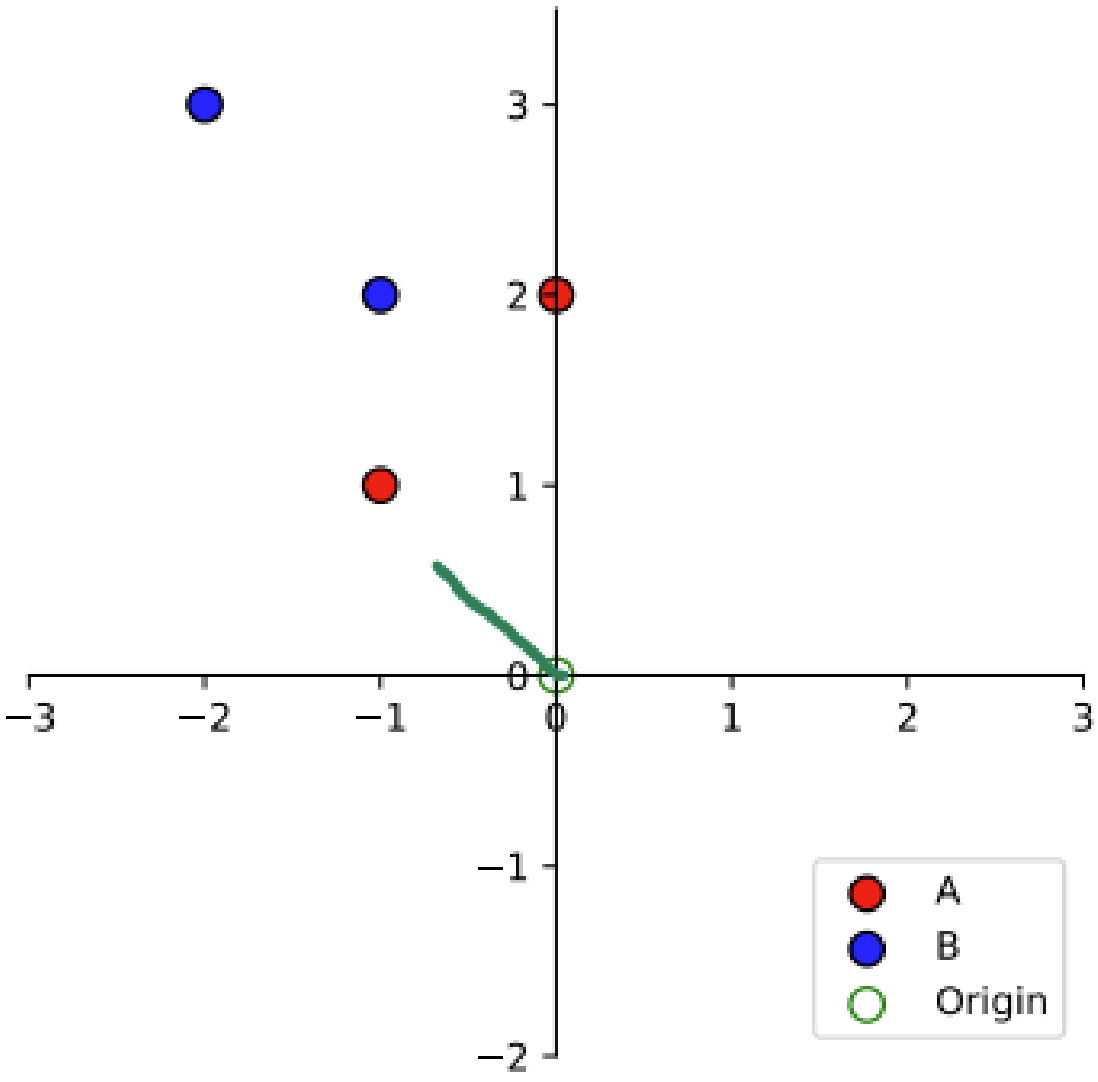


Figure 1: PCA

1b

No it doesn't work, consider the example... $W^T = \begin{bmatrix}1 & 1\end{bmatrix}$ and $x^t = \begin{bmatrix}1\\0\end{bmatrix}$ then

$$z^t = W^T x^t$$

$$z^t = \begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix}$$

$$z^t = \begin{bmatrix}1\end{bmatrix}$$

So now we've used W^T to project the 2-dimensional data to 1-dimensional data. However, we see that when we use the PCA projection matrix again, it does not return to the original data.

$$v^t = W z^t$$

$$v^t = \begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1\end{bmatrix}$$

$$v^t = \begin{bmatrix}1\\1\end{bmatrix} \neq x^t$$

1c

The Within-Class Scatter Matrix is $S_W = \sum_{i=1}^{K}\sum_t r_i^t (x^t - m_i)(x^t - m_i)^T$

$$S_W = \sum_t (x^t - m_1)(x^t - m_1)^T + \sum_t (x^t - m_2)(x^t - m_2)^T$$

We can eliminate the r_i^t term once we split into summations by class.

$$m_1 = \begin{bmatrix}-0.5\\1.5\end{bmatrix}$$

$$m_2 \begin{bmatrix}-1.5\\2.5\end{bmatrix}$$

$$S_W = \begin{bmatrix}0.5\\0.5\end{bmatrix}\begin{bmatrix}0.5\\0.5\end{bmatrix}^T + \begin{bmatrix}-0.5\\-0.5\end{bmatrix}\begin{bmatrix}-0.5\\-0.5\end{bmatrix}^T + \begin{bmatrix}0.5\\-0.5\end{bmatrix}\begin{bmatrix}0.5\\-0.5\end{bmatrix}^T + \begin{bmatrix}-0.5\\0.5\end{bmatrix}\begin{bmatrix}-0.5\\0.5\end{bmatrix}^T$$

$$S_W = \begin{bmatrix}0.25 & 0.25\\0.25 & 0.25\end{bmatrix} + \begin{bmatrix}0.25 & 0.25\\0.25 & 0.25\end{bmatrix} + \begin{bmatrix}0.25 & -0.25\\-0.25 & 0.25\end{bmatrix} + \begin{bmatrix}0.25 & -0.25\\-0.25 & 0.25\end{bmatrix}$$

$$S_W = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$$

The Between-Class Scatter Matrix is $S_B = (m_1 - m_2)(m_1 - m_2)^T$

$$S_B = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix}$$

$$S_B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

1d

By theorem (6.46) in Alpaydin, $w = S_W^{-1}(m_1 - m_2)$

$$w = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$w = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

After we normalize w, we get

$$w = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{bmatrix}$$

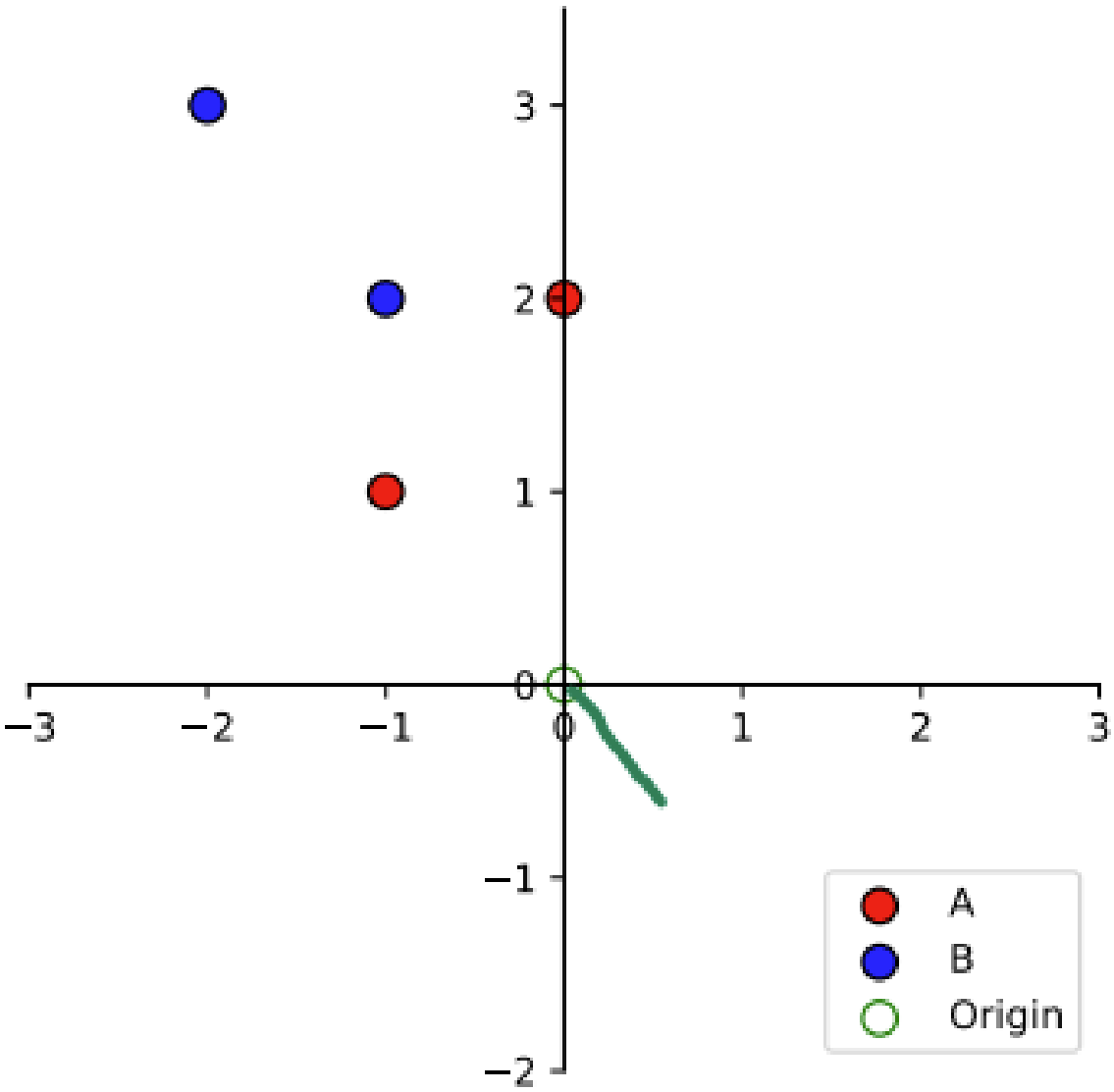


Figure 2: LDA

Problem 2

$x_1 = 1, x_2 = 4, x_3 = 5, x_4 = 6, x_5 = 7, x_6 = 8, x_7 = 10, x_8 = 12, x_9 = 14$
$k = 3$

2a

Use $c_1 = 0, c_2 = 5, c_3 = 10$
Iteration 1 cluster assignment:
$cluster_1 = \{x_1\}$
$cluster_2 = \{x_2, x_3, x_4, x_5\}$
$cluster_3 = \{x_6, x_7, x_8, x_9\}$
Iteration 1 update means: $c_1 = 1, c_2 = 5.5, c_3 = 11$
Iteration 2 cluster assignment:
$cluster_1 = \{x_1\}$
$cluster_2 = \{x_2, x_3, x_4, x_5, x_6\}$
$cluster_3 = \{x_7, x_8, x_9\}$
Iteration 2 update means: $c_1 = 1, c_2 = 6, c_3 = 12$
Iteration 3 cluster assignment:
$cluster_1 = \{x_1\}$
$cluster_2 = \{x_2, x_3, x_4, x_5, x_6\}$
$cluster_3 = \{x_7, x_8, x_9\}$

2b

By the third iteration, no changes were made to cluster assignments, so it took three iterations for the k-means algorithm to converge.
$J = \sum_{n=1}^{N} \sum_{k=1}^{K} r_{nk} ||X_n - \nu_k||^2$
$J = (1-1)^2+(4-6)^2+(5-6)^2+(6-6)^2+(7-6)^2+(8-6)^2+(10-12)^2+(12-12)^2+(14-12)^2$
$J = 18$
The distortion measure is 18.

2c

Use $c_1 = 2, c_2 = 7, c_3 = 12$
Iteration 1 cluster assignment:

$cluster_1 = \{x_1, x_2\}$
$cluster_2 = \{x_3, x_4, x_5, x_6\}$
$cluster_3 = \{x_7, x_8, x_9\}$
Iteration 1 update means: $c_1 = 2.5, c_2 = 6.5, c_3 = 12$
Iteration 2 cluster assignment:
$cluster_1 = \{x_1, x_2\}$
$cluster_2 = \{x_3, x_4, x_5, x_6\}$
$cluster_3 = \{x_7, x_8, x_9\}$

2d

The k-means algorithm with new cluster centers took two iterations to reach convergence.
$J = \sum_{n=1}^{N} \sum_{k=1}^{K} r_{nk} ||X_n - \nu_k||^2$
$J = (1-2.5)^2 + (4-2.5)^2 + (5-6.5)^2 + (6-6.5)^2 + (7-6.5)^2 + (8-6.5)^2 + (10-12)^2 + (12-12)^2 + (14-12)^2$
$J = 17.5$
The distortion measure is 17.5.

2e

The second solution (c) is better. Not only did it converge in fewer iterations, but it also resulted in a lower distortion measure.

Problem 3

3a

Report the number of iterations for convergence and the classification accuracy on the test samples.

Using the raw data, it took 27 iterations and achieved a classification accuracy of 0.94.

```
Using raw data converged in 27 iteration (3.06 seconds)
Classification accuracy: 0.94
##################
Project data into 73 dimensions with PCA converged in 26 iteration (2.45 seconds)
Classification accuracy: 0.94
##################
Project data into 1 dimension with PCA converged in 33 iteration (2.84 seconds)
Classification accuracy: 0.74
```

Figure 3: results

Plot the history of reconstruction errors.

Please see the appendix for plots.

Is the plot shape following what you expect?

The overall shape of the three plots follow what is expected of k-means clustering. As iterations progress, the error in each iteration monotonically decreases which shows k-means working. What's unexpected is the "bump" in each graph that shows the error rate still falling, but slowing down before speeding up again.

3b

How many dimensions are necessary in this case?

73 dimensions were necessary to maintain 90% of the data.

Does PCA help clustering? When we use 73 dimensions instead of the full 784, runtime improves by 0.61 seconds and it takes one less iteration to converge. Meanwhile, the classification accuracy remains the same at 94%. This shows that using principle component analysis to reduce dimensionality while maintaining 90% of the data will help k-means cluster by improving speed.

3c

When we reduce dimensionality to 1 dimension, we lose a lot of data and classification accuracy takes a big decrease down from 94% to 74%. This means that the results are worse. The benefits of dimensionality reduction, like reduced runtime, are also not seen here with 1 dimension. Operating with 1 dimension, it actually took longer and more iterations to reach convergence. This shows that the best number of dimensions is not necessarily the lowest.

Appendix

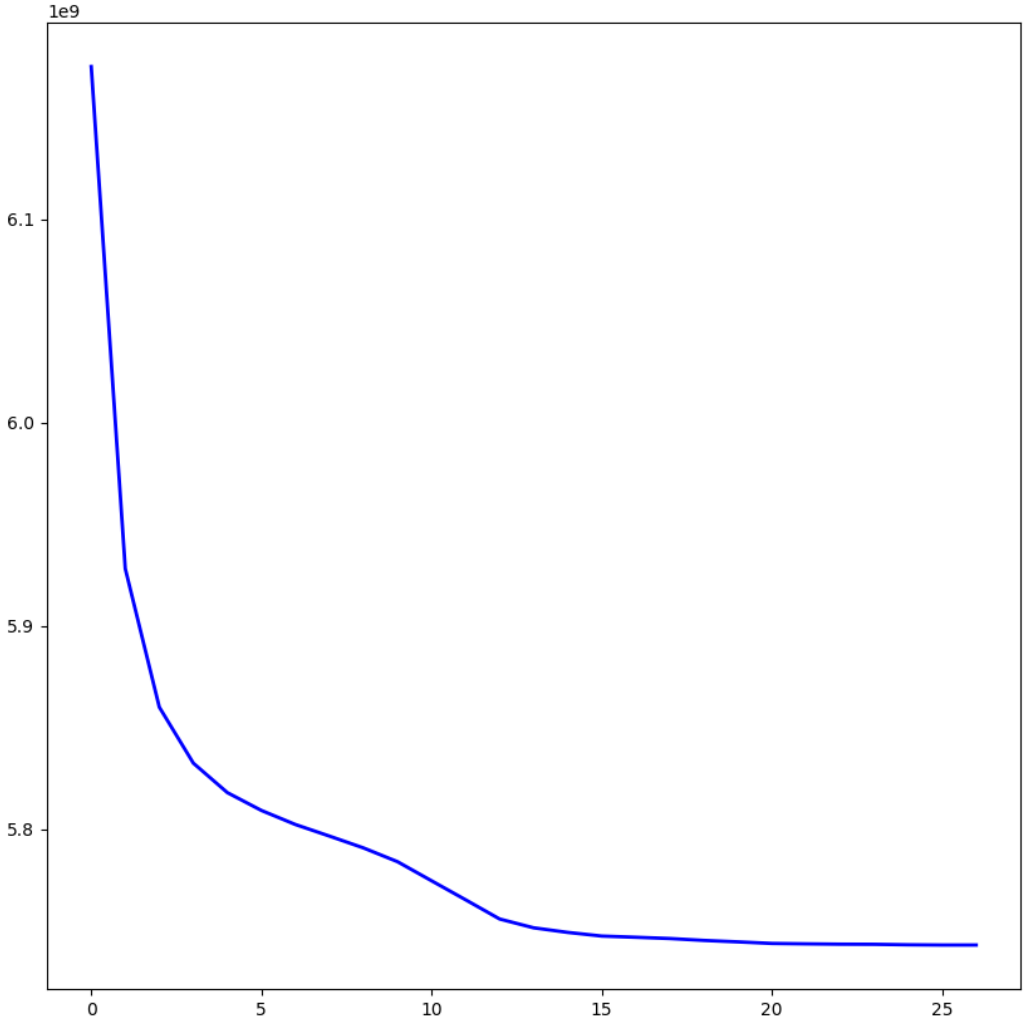


Figure 4: raw data

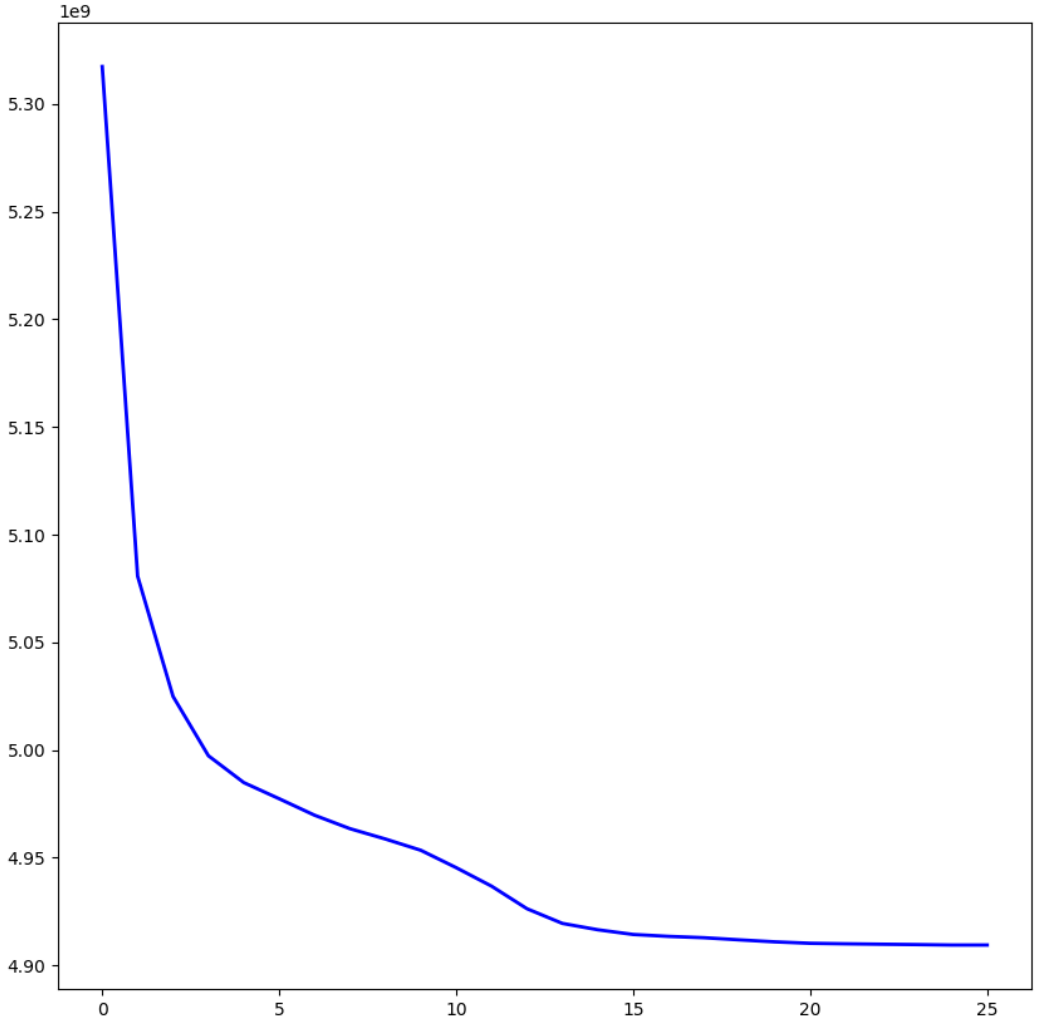


Figure 5: pca

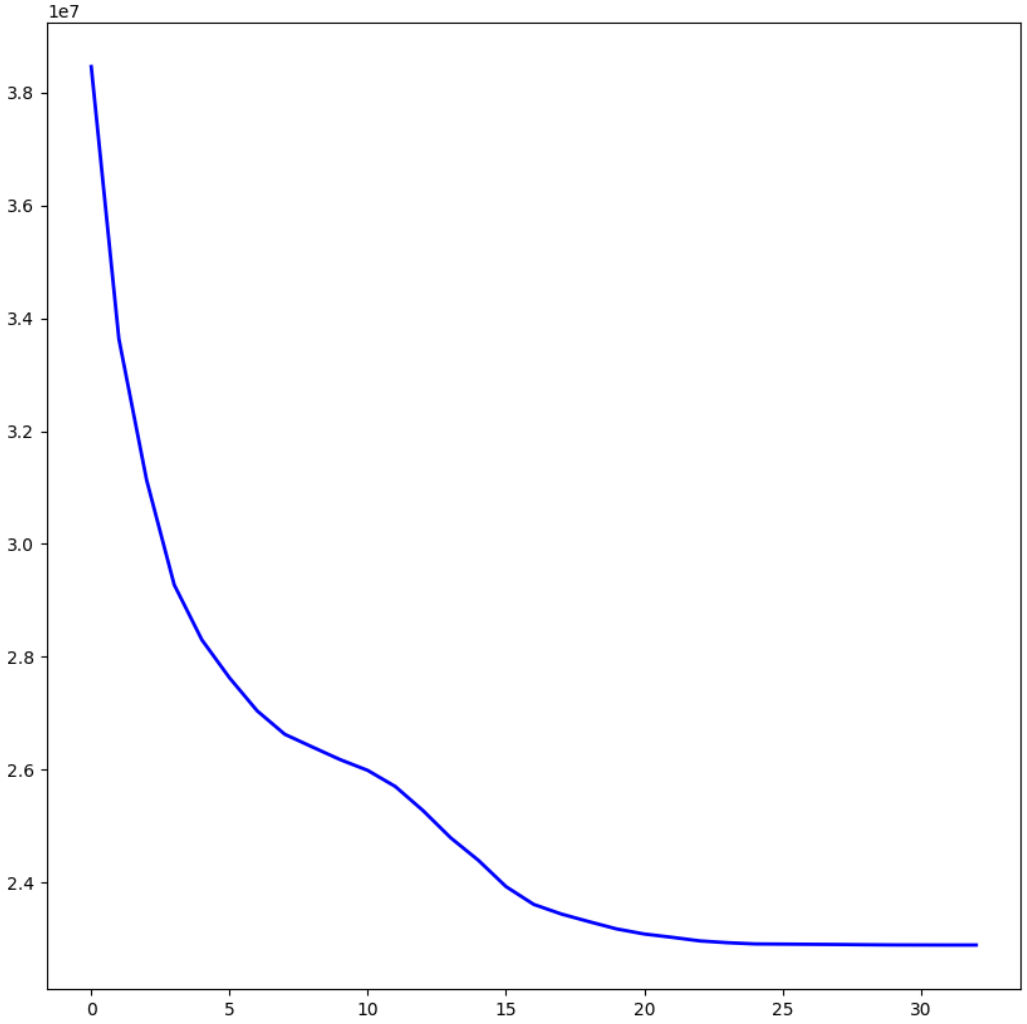


Figure 6: pca with one dimension